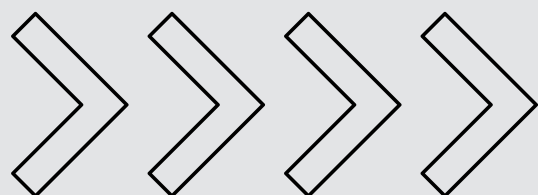


অলীকবচন(OLIKBOCHON)

**Adaptive Multi-Signal LLM-Judge Ensembling
with Verified Deterministic Retrieval for Bengali Hallucination Detection**



Presentation by DU_Outliers

The Task Defined

01 Three inputs, one binary label

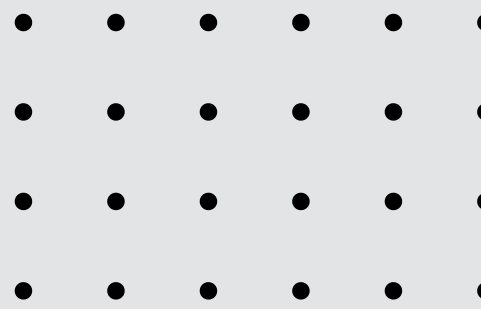
Bengali prompt + candidate response + optional context passage → faithful or hallucinated. Scored as F1 on the hallucinated class only.

02 Three cultural-distance bands

Globally stable facts · Bangladesh-specific knowledge · contested or recent facts. The failure mode concentrates in the middle band — exactly where general-purpose LLMs are weakest.

03 Importance of Context

299 labeled rows for local validation. Nothing to fine-tune on. So we treated this as a calibration problem, not a training problem.



INTRODUCTION

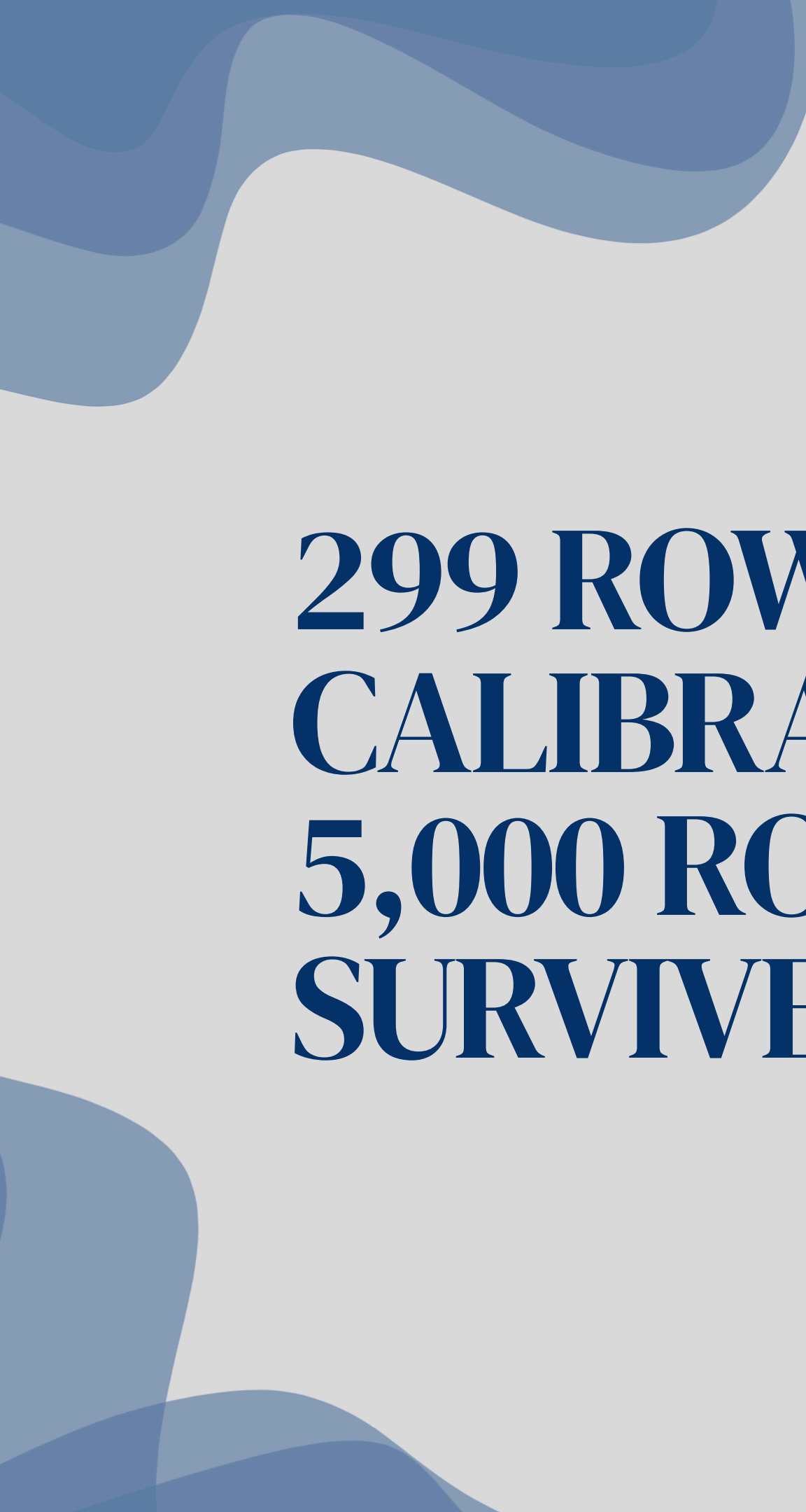
280M+ Bengali speakers. Almost no hallucination-detection research.

Because fluent Bengali can be confidently **wrong—and still look completely right.**

অলীকবচন challenges every answer through

one local judge, three specialist routes, and one rule: evidence before intuition.

F1: 0.90216 · 6th among finalists



**299 ROWS TO
CALIBRATE.
5,000 ROWS TO
SURVIVE.**

Our cross-validated numbers are optimistic, lower-variance estimates — not predictions of held-out performance. We designed around that instead of pretending otherwise.



Route

Identify the query and direct it to relevant pathways.

Attach Evidence

Integrate supporting data and context for informed decisions.

Safe Override

Apply checks to ensure accuracy before reaching a verdict.

Judge

Evaluate outcomes and determine final responses based on evidence.

PROBLEM

Fluent does not mean faithful. For example:- A model names a poem's author correctly in English — and misattributes it in Bengali.

**Bengali
Question**

+

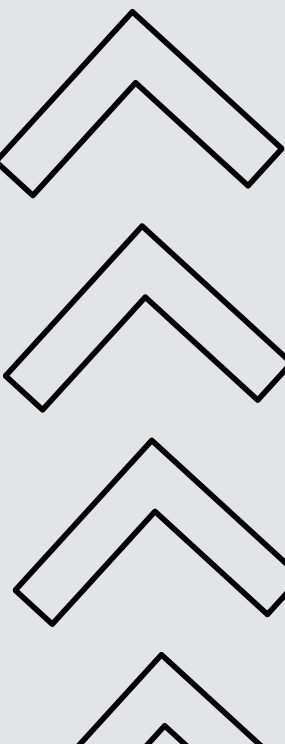
**Candidate
Response**

+

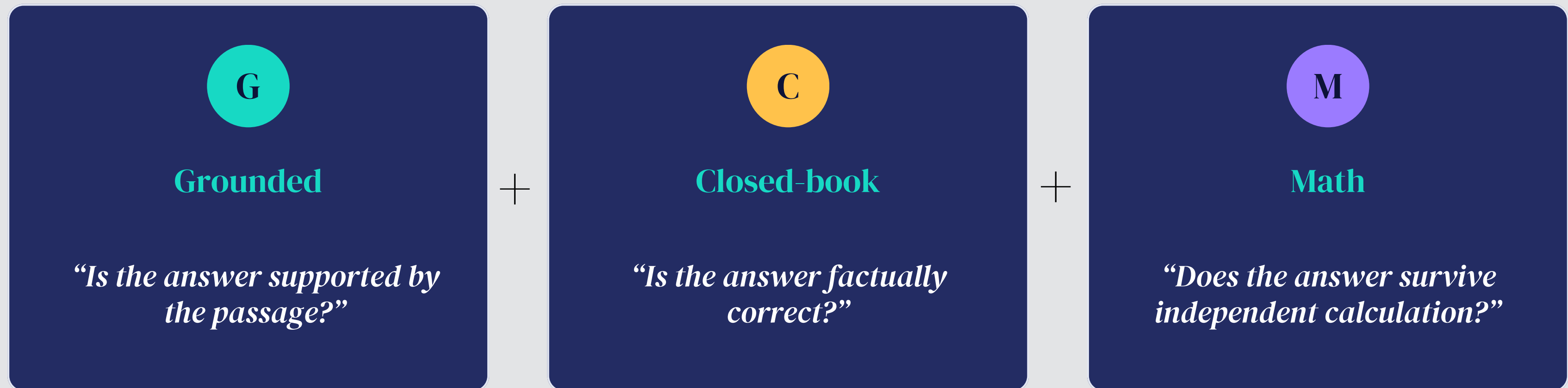
**Optional
Context**



Faithfil or Hallucinated?

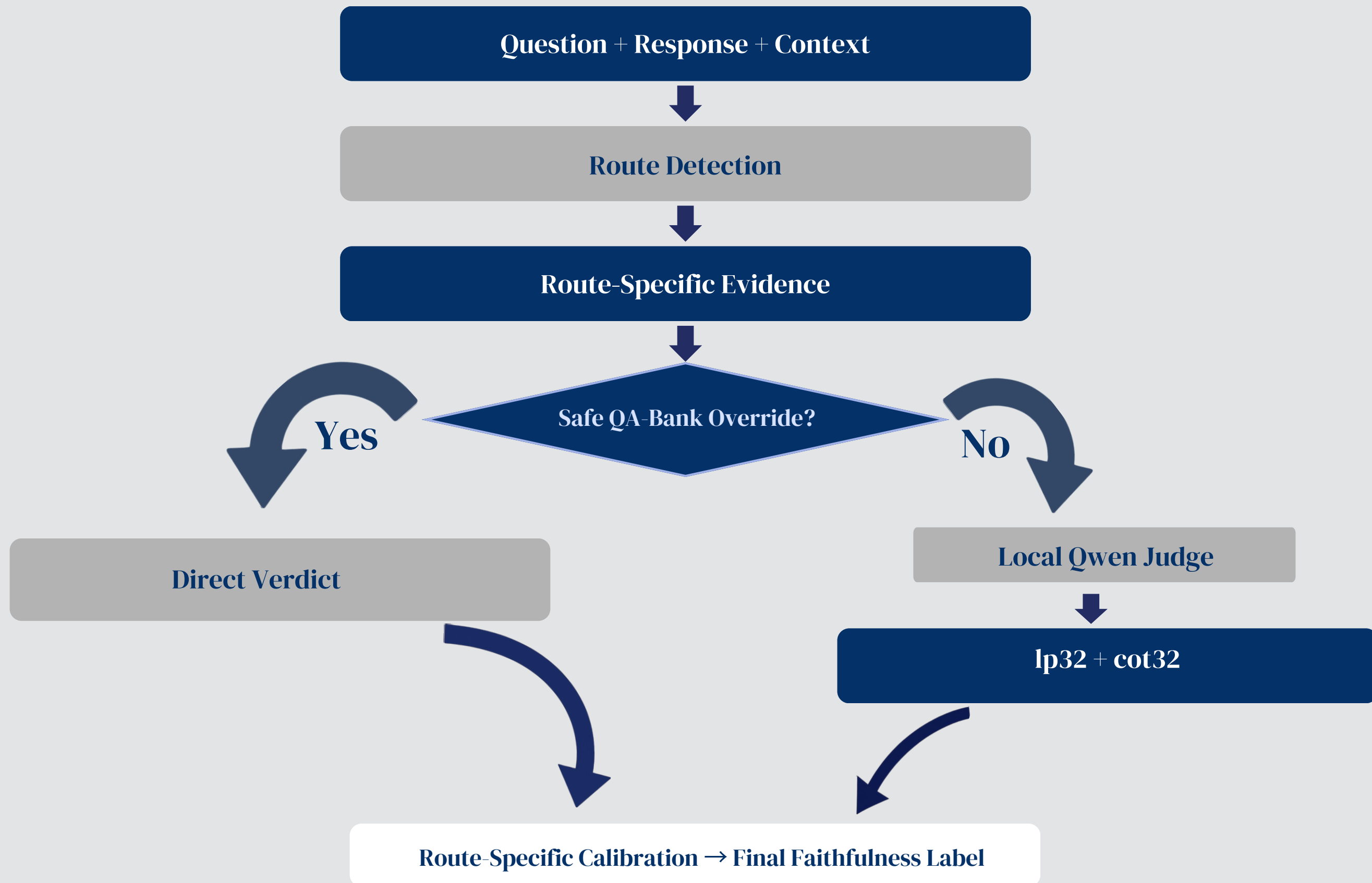


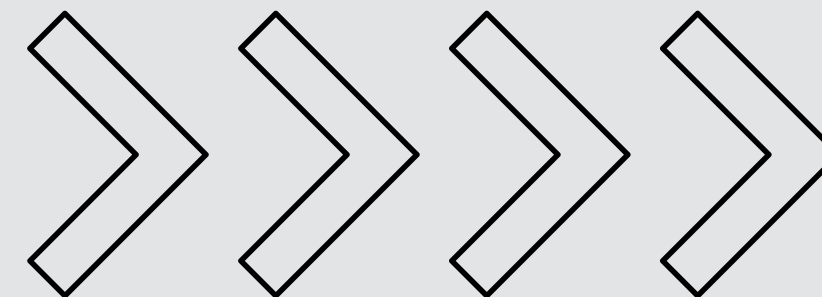
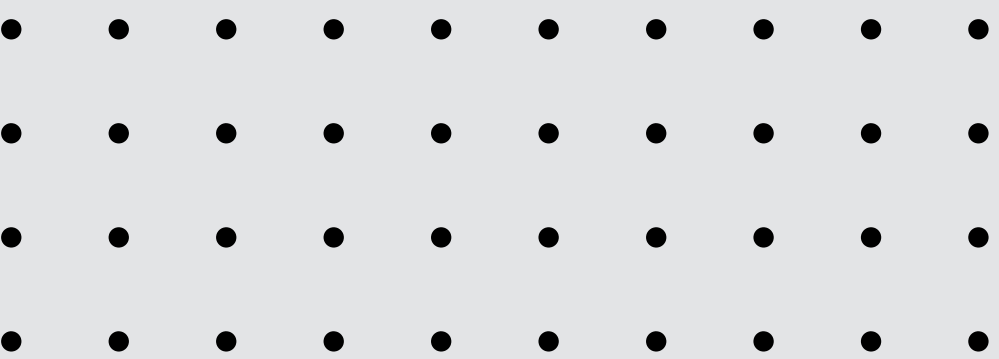
Different questions hallucinate differently.



One model. Three specialist verification routes.

Our Solution





RIGHT STRATEGY



Grounded

“Is it supported?”

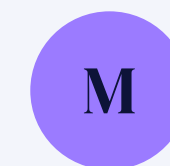
- Reads the provided context
- Checks the response answers the actual question
- Verifies names, dates, numbers, entities
- Rejects unsupported claims despite similar wording



Closed-book

“Can we verify it?”

- Retrieves Bengali Wikipedia evidence via BM25
- Searches curated Bengali QA & exam banks
- Combines evidence with judge knowledge
- Uses cautious deterministic overrides



Math

“Can we reproduce it?”

- Solves the problem independently
- Compares derived vs. proposed answers
- Accepts equivalent forms (fractions, decimals)

EVIDENCE

We do not ask the judge to guess when evidence exists.

Dataset

- Bengali-aware BM25 retrieval
- Bengali Wikipedia + public QA banks
- Bengali digit normalization
- QA sources indexed before Wikipedia
- Safe abstention when a match is uncertain

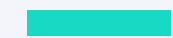
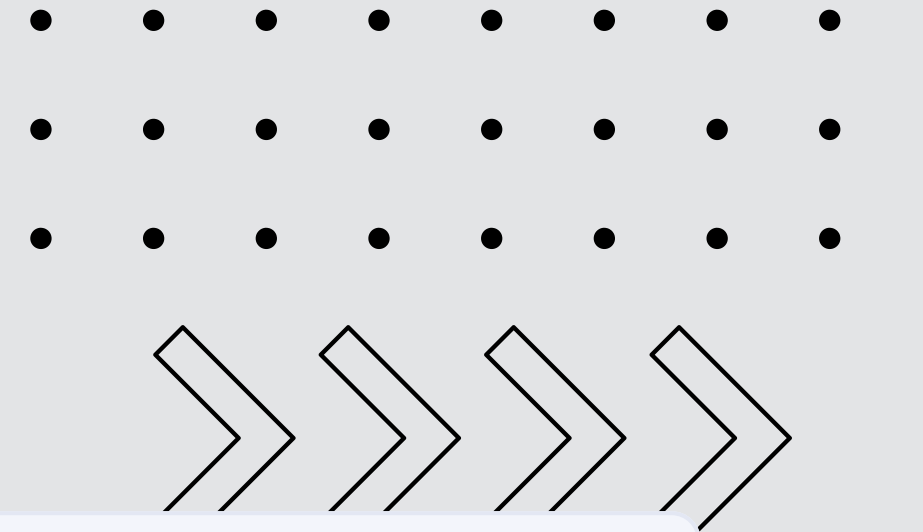
QA-Bank Protection Rules

- › Question coverage $\geq 80\%$
- › Response must appear in the matched options
- › Negative override only when similarity ≥ 0.95

500K → 900k passages

Retrieval corpus capacity expanded for broader Bengali coverage

CONFIDENCE AND EVIDENCE DOES NOT AGREE



Signal 1 — lp32

Single-token Yes/No probability from the judge.



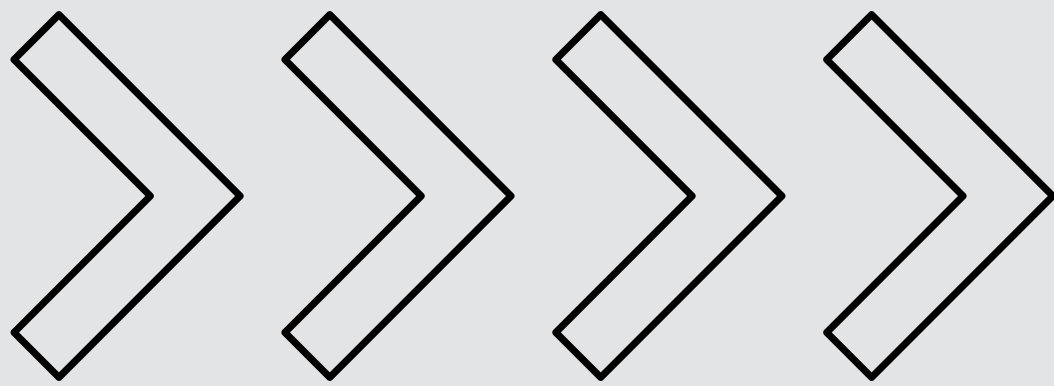
Signal 2 — cot32

Multiple independent reasoning attempts, then a self-consistency vote.

$$\text{Score}(\text{route}) = \textcolor{teal}{w}_1 \times \text{lp32} + \textcolor{orange}{w}_2 \times \text{cot32}$$

- Separate weights for each route
- Separate threshold for each route
- Selected through 5-fold cross-validation





STRENGTH

Our strongest feature was knowing what to turn off.



1 • Question-template collisions

Similar QA-bank questions could refer to a different entity.

0.858 → 0.838 leaderboard drop → asymmetric matching + abstain on mismatch



2 • LoRA calibration leakage

Verifier trained on the same 299 rows later used for calibration.

Scores of 0.767 / 0.806 → LoRA disabled until truly held-out data exists



3 • Stale dense-retrieval cache

E5 embeddings validated only by row count; corpus order changed silently.

Hybrid retrieval disabled — reliable BM25 retained



4 • CV vs. leaderboard gap

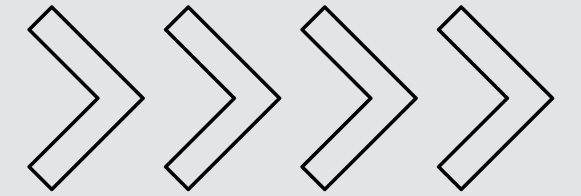
299 rows did not fully represent the hidden test distribution.

5-fold CV kept competition-only; extended calibration never inflates it

DEADLINE

A powerful pipeline is useless
if it misses the deadline.

- Qwen2.5-32B-Instruct-AWQ primary judge, Qwen2.5-14B fallback
- Fully local inference through vLLM, 4-bit AWQ quantization
- Dual-T4 execution, under the 50GB model limit
- Automatically reduces reasoning votes for over 4,000 rows
- Same notebook supports online and offline execution



3–5 hrs

estimated full runtime

9 hrs

competition limit

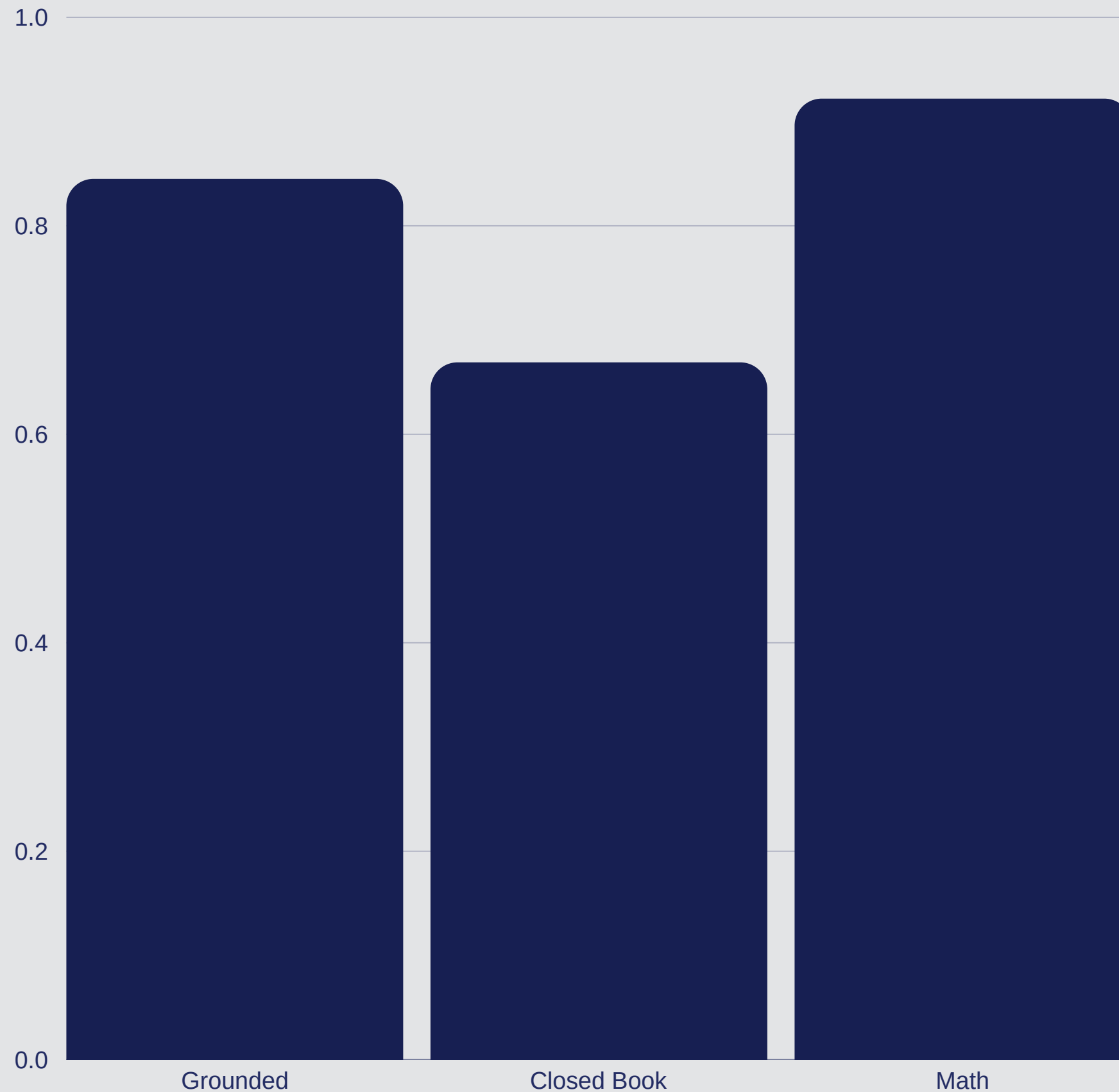
1.5–2.5 hrs

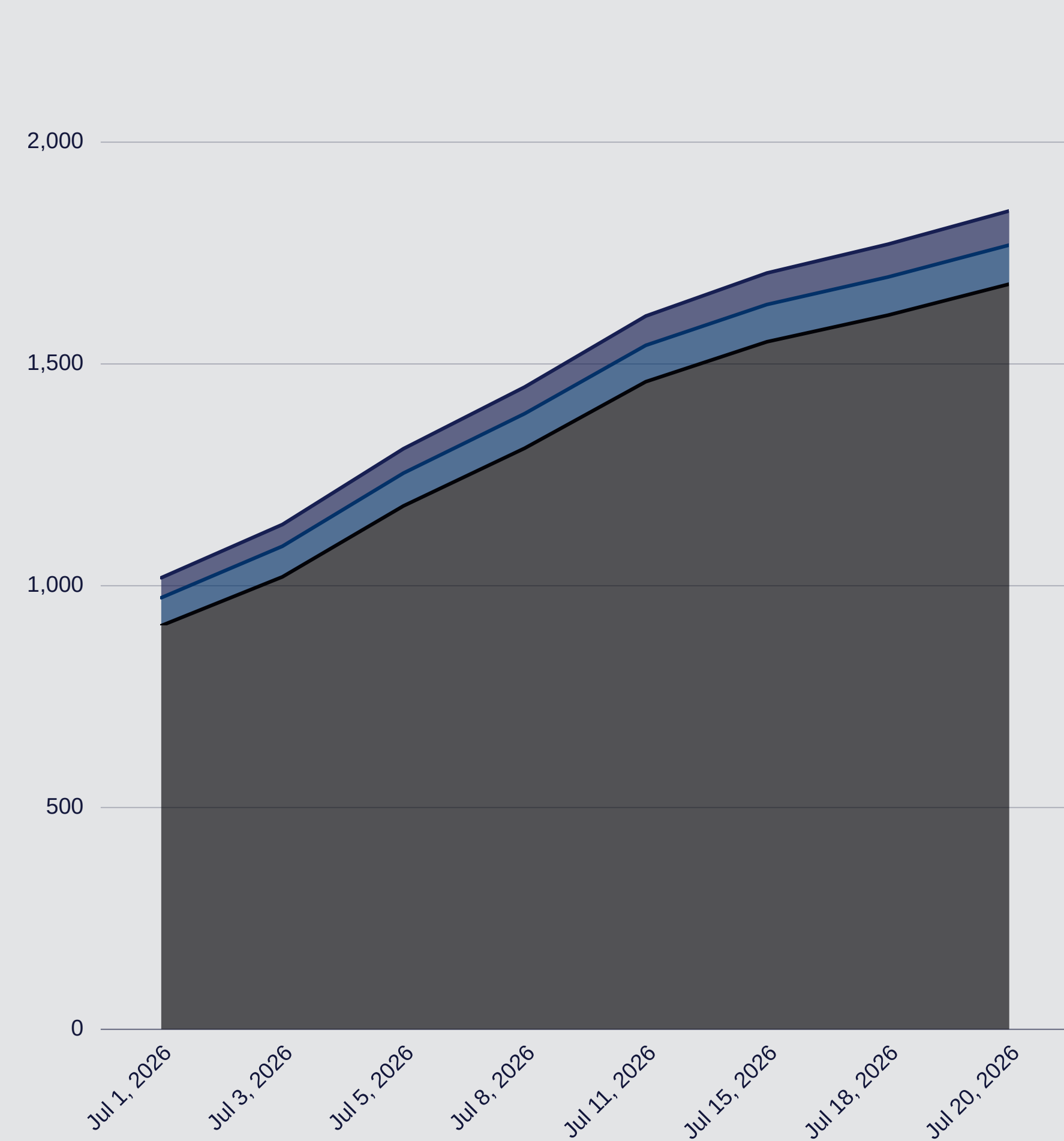
warm cached reruns



Calibration Results

This chart illustrates our F1 scores across three routes: Grounded, Closed Book, and Math. Notably, the Math route achieved the highest score, indicating its effectiveness in our detection system's performance.



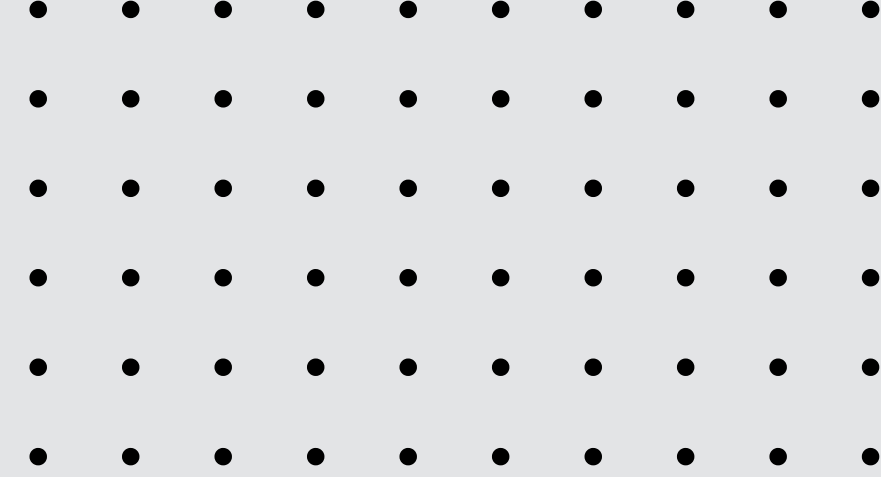
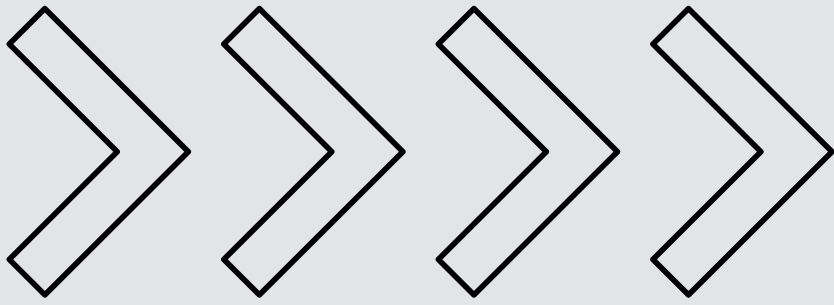


Runtime Engineering Insights

The chart highlights performance fluctuations during the nine-hour runtime, demonstrating significant improvements with dual T4 GPUs, revealing a clear correlation between resource allocation and processing efficiency.

More trustworthy by design — not merely more complex.

Common Approach	অলীকবচন
One prompt for every question	Three specialized routes
Judge-only factual decisions	Retrieved Bengali evidence
One generated verdict	Dual-signal scoring
One global threshold	Per-route calibration
Always trust retrieval matches	Options verification and abstention
Add every available component	Evidence-backed ablation
Hosted external APIs	Fully local and reproducible
Maximum computation everywhere	Adaptive computation

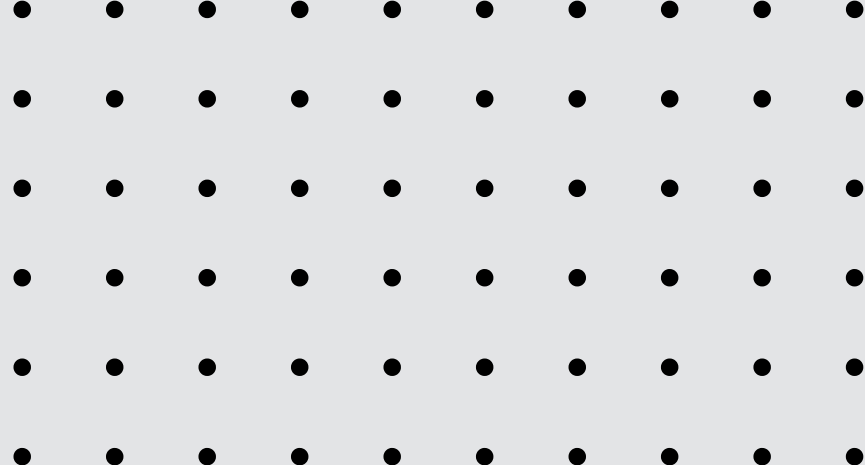


SIGNING OFF

Trust is not a bigger model.

Turning confident Bengali answers into accountable decisions.

It is better evidence, better routing, and better decisions.



THANK YOU

For any queries , feel free to ask..

